

How Much Area Does a Circle

Force?

Gromov's filling-area conjecture through distance profiles and Fourier modes

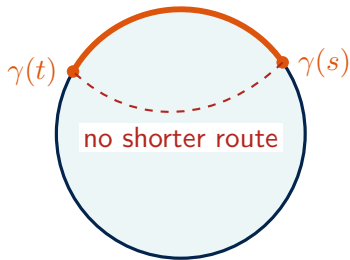
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Joint work with Xiaolong Li and Yimin Zhong

Mother deck — September 2026

A circle makes a contract with its filling



boundary distances must survive

Isometric filling

A compact connected Riemannian surface M with one boundary circle satisfies

$$d_M(\gamma(s), \gamma(t)) = d_{\mathbb{T}}(s, t) \quad (s, t \in \mathbb{T}).$$

- ▶ Walking along the shorter boundary arc already gives " $\leq$ ".
- ▶ The content is " $\geq$ ": the interior creates **no shortcut**.
- ▶ The unit hemisphere honors the contract and has area 2π .

Source: Companion manuscript, Introduction, lines 115–148.

Gromov's conjecture: the hemisphere should be cheapest

Filling-area conjecture (Gromov, 1983)

Every compact **orientable** Riemannian isometric filling of the circle of circumference 2π should satisfy

$$\text{Area}(M) \geq 2\pi,$$

with equality for the unit round hemisphere.

KNOWN disk case
Gromov's theorem

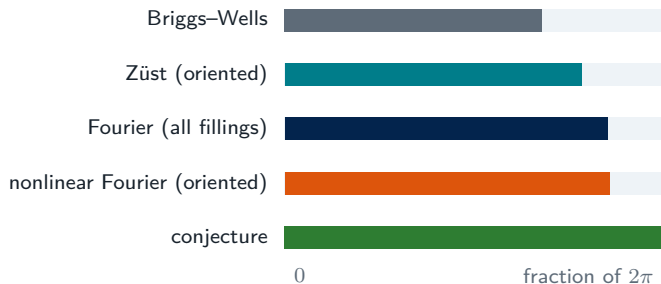
KNOWN genus one
Bangert–Croke–Ivanov–Katz

OPEN general genus
This is the conjecture

The theorem in this deck improves global lower bounds; it does not prove the conjectural value 2π .

Source: Gromov (1983); Bangert–Croke–Ivanov–Katz (2005); companion manuscript, lines 140–153.

A quantitative scoreboard

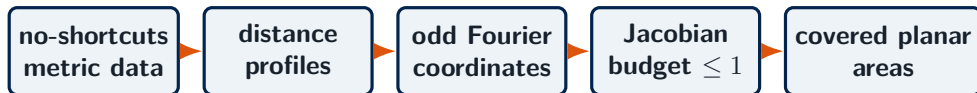


selected bound	value
$\frac{\sqrt{3}}{4}\pi^2$	4.27366
$\frac{\pi^2}{2}$	4.93480
$\frac{14\zeta(3)}{\pi}$	5.35677
nonlinear	> 5.38982
conjecture	6.28319

The last step is numerically modest but conceptually new: it is the first gain here that uses nonlinear relations among Fourier modes.

Source: Companion manuscript, lines 155–214. Values are rounded only on this overview frame.

The proof pipeline: one budget, many planar payoffs



Analysis supplies the budget

Lipschitz functions + Rademacher + Bessel control how much total two-dimensional Jacobian the coordinates can spend.

Topology certifies the payoff

Boundary winding is converted into regions that every extension must cover, even when the filling is nonorientable.

Source: Companion manuscript, "Structure of the argument," lines 216–257.

Distance profiles turn geometry into functions

View the boundary from x

For every boundary label θ ,

$$u_\theta(x) = d_M(x, \gamma(\theta)).$$

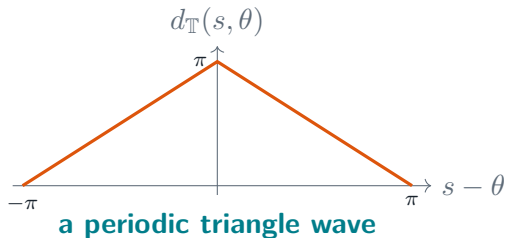
Each u_θ is 1-Lipschitz in x .

On the boundary, no shortcuts determine the whole profile:

$$u_\theta(\gamma(s)) = d_{\mathbb{T}}(s, \theta).$$

The unknown interior geometry has acquired **known boundary data**.

Source: Companion manuscript, Section 2, lines 259–314.



Antipodes split the profile into signal and slack

$$a_\theta(x) = \frac{u_\theta(x) - u_{\theta+\pi}(x)}{2}, \quad a_{\theta+\pi} = -a_\theta,$$
$$\sigma_\theta(x) = \frac{u_\theta(x) + u_{\theta+\pi}(x) - \pi}{2}, \quad \sigma_\theta \geq 0.$$

one unit
of energy

odd signal a_θ
survives in odd Fourier modes

slack σ_θ
measures unused antipodal equality

Almost-everywhere energy identity

$$|\nabla a_\theta|^2 + |\nabla \sigma_\theta|^2 = 1.$$

On the boundary, $\sigma_\theta = 0$: all boundary energy is signal.

Source: Companion manuscript, Definition 2.1 and equation (2.8), lines 279–393.

The Fourier camera sees exact boundary loops

For odd $n \geq 1$, define a planar coordinate

$$z_n(x) = A_n(x) + iB_n(x),$$

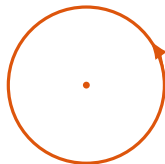
$$z_n(x) = \frac{1}{\pi} \int_0^{2\pi} a_\theta(x) e^{in\theta} d\theta.$$

Anti-periodicity kills every even mode.

Boundary miracle

$$z_n(\gamma(s)) = -\frac{4}{\pi n^2} e^{ins}.$$

Source: Companion manuscript, Sections 3–4, lines 394–434 and 486–523.



$n = 1$
winds 1 time
radius $4/(\pi 1^2)$



$n = 3$
winds 3 times
radius $4/(\pi 3^2)$



$n = 5$
winds 5 times
radius $4/(\pi 5^2)$

Bessel's inequality gives one shared Jacobian budget

Select finitely many odd modes and mix them by any orthogonal matrix $U \in O(N)$:

$$G_j = \sum_{k=1}^N u_{jk} F_{m_k}, \quad F_n = (A_n, B_n).$$

At almost every point of M ,

$$\sum_{j=1}^N J_2 G_j \leq 1.$$

Orthogonal mixing redistributes the budget; it does not create new derivative energy.

Source: Companion manuscript, Proposition 3.2, lines 415–485.

one unit of local area budget



each coordinate spends a nonnegative share of the same Bessel energy

Sharper version

The unused slack energy is subtracted from the right-hand side.

Why nonorientability breaks the obvious winding argument

On an oriented filling

A signed Jacobian and Stokes' theorem can read the integer winding of each boundary loop.

winding $n \implies$ signed area payoff.

SIGN CONSISTENT

On a nonorientable filling

A continuously chosen area sign does not exist. Local orientation can flip, so signed contributions cannot simply be added.

n -fold loop $\not\Rightarrow$ n signed sheets.

SIGN CAN FLIP

Repair strategy: turn each multiple loop into a Jordan curve, then count coverage modulo 2.

Source: Companion manuscript, Section 5, especially Lemmas 5.3–5.4.

Untwisting: orthogonal mixing makes every row simple

Start with boundary modes of frequencies

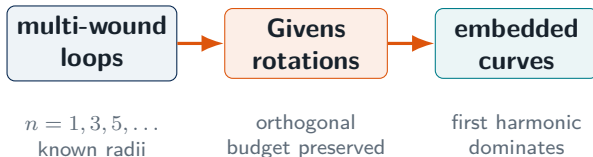
$$1, 3, 5, \dots, 2N - 1.$$

An explicit product of Givens rotations produces $U_N \in O(N)$ whose every row satisfies

$$|u_{j1}| > \sum_{k=2}^N \frac{|u_{jk}|}{2k-1}.$$

The first harmonic outruns the combined derivative of all higher harmonics.

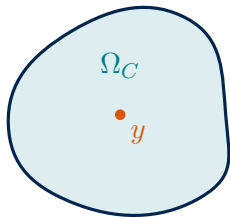
Source: Companion manuscript, Lemmas 5.1–5.2, lines 524–636.



The design constraint

Change the boundary geometry without changing the total Jacobian budget.

Jordan curves turn topology into guaranteed area



odd boundary degree forbids
omitting an interior point

Two facts do the work

1. First-harmonic dominance makes $G_j|_{\partial M}$ a Jordan curve C_j with explicit enclosed area.
2. Mod-2 degree forces the bounded component $\Omega_{C_j} \subseteq G_j(M)$.

Hence the area formula gives

$$\int_M J_2 G_j \, dA \geq |\Omega_{C_j}|.$$

No global orientation is needed: parity survives sign changes.

Add the payoffs, spend the budget once

$$\text{Area}(M) \geq \sum_{j=1}^N \int_M J_2 G_j \, dA \quad (\text{one shared budget})$$

$$\geq \sum_{j=1}^N |\Omega_j| \quad (\text{forced coverage})$$

$$= \frac{16}{\pi} \sum_{k=1}^N \frac{1}{(2k-1)^3} \quad (\text{orthogonality}).$$

Let $N \rightarrow \infty$ and use

$$\sum_{k \geq 1} \frac{1}{(2k-1)^3} = \frac{7}{8} \zeta(3).$$

Source: Companion manuscript, Proposition 5.5 and Theorem 1.1, lines 709–748; exact Lean interface in FORMALIZATION.md.

$$\begin{aligned} \text{Area}(M) &\geq \frac{14\zeta(3)}{\pi} \\ &= 5.3567723444\dots \\ &\quad \text{orientation-free} \end{aligned}$$

THEOREM

The conjectural target is still 2π .

The discarded slack becomes an area defect

Antipodal defect refinement

Every isometric filling, orientable or not, satisfies

$$\text{Area}(M) \geq \frac{14\zeta(3)}{\pi} + \frac{1}{2\pi} \int_M \int_0^{2\pi} |\nabla \sigma_\theta(x)|^2 d\theta dA(x).$$

If slack varies

The baseline improves automatically by a nonnegative Dirichlet energy.

If the defect vanishes

The profile lies infinitesimally in the antipodal geometry of the injective hull; a different idea is needed.

A good estimate is also a diagnostic: it tells us where improvement must hide.

Source: Companion manuscript, Theorem 6.1, lines 749–793.

Oriented area can be read by a calibration

Gather all odd coefficients into

$$\mathcal{H} = \ell^2(1, 3, 5, \dots; \mathbb{C}), \quad Z : M \longrightarrow \mathcal{H},$$

$$Z(x) = (z_1(x), z_3(x), z_5(x), \dots).$$

The standard symplectic form

$$\omega(p, q) = \sum_{n \text{ odd} > 0} \operatorname{Im}(\overline{p_n} q_n)$$

has comass at most 1 on admissible tangent planes.

Source: Companion manuscript, Section 7; exact status boundary in the slide claim ledger.

Area detector

$$\begin{aligned} \int_{\partial M} Z^* \alpha &= \int_M Z^* \omega && \text{(Stokes)} \\ &\leq \operatorname{Area}(M) && (\operatorname{comass} \leq 1). \end{aligned}$$

The left side is known from the boundary loop.

The boundary action equals $14\zeta(3)/\pi$, recovering the linear bound in the oriented setting.

Two natural nonlinear ideas hit the same wall

Scalar feature engineering

Replacing each distance by a separate 1-Lipschitz feature $\psi(u_\theta)$ cannot improve

$$\frac{14\zeta(3)}{\pi}.$$

A positive logarithmic kernel makes the original choice extremal.

Ambient Hilbert geometry

The boundary Fourier loop spans a holomorphic disk whose Euclidean area is already

$$\frac{14\zeta(3)}{\pi}.$$

No Euclidean-comass-1 form can read more action.

To gain, couple modes genuinely nonlinearly and exploit the smaller L^∞ geometry of actual distance profiles.

Source: Companion manuscript, Propositions 8.1–8.2, lines 905–1032. Each barrier is limited to its stated class.

Phase resonance: a cubic term lands on the right frequency

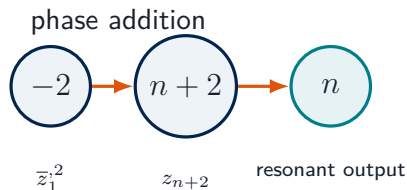
For $n = 1, 3, 5, \dots$, perturb the coordinate by

$$(\mathcal{G}_\lambda(z))_n = z_n + \lambda \overline{z_1}^2 z_{n+2}.$$

On the boundary,

$$\overline{e^{is}}^2 e^{i(n+2)s} = e^{ins}.$$

The cubic correction has **exactly the same phase** as z_n .



Boundary payoff

$$\mathcal{B}(\lambda) = C_0 + b\lambda + d\lambda^2 \text{ with } b > 0.$$

Source: Companion manuscript, Sections 9–10, lines 1033–1141.

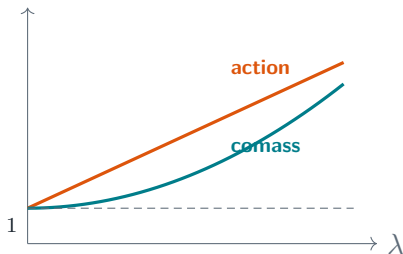
The key asymmetry: gain is linear, cost is quadratic

At a plane saturating the linear comass, the boundary-direction field is analytic or anti-analytic with $|W| = 1$. Therefore its nonzero autocorrelation vanishes:

$$\sum_{\substack{n>0 \\ n \text{ odd}}} c_n \overline{c_{n+2}} = 0.$$

This kills the first variation of the comass.

$$\begin{array}{l} \text{boundary gain} = O(\lambda), \\ \text{comass cost} = O(\lambda^2). \end{array}$$



The picture is schematic; the theorem uses explicit global constants.

Source: Companion manuscript, Sections 10–11; exact status boundary in the slide claim ledger.

An explicit nonlinear certificate

For $0 \leq \lambda < \pi^2/32$,

$$\text{Area}(M) \geq \frac{\mathcal{B}(\lambda)}{\mathcal{C}(\lambda)} = \frac{C_0 + b\lambda + d\lambda^2}{1 + \lambda^2 \left[Q_* + \frac{C_*^2}{4(1 - D_*\lambda)} \right]}.$$

At $\lambda = 0.03$,

$$\mathcal{B} = 5.4230028938\dots, \quad \mathcal{C} = 1.0061557533\dots$$

$\text{Area}(M) > 5.38982446$
every compact connected oriented
Riemannian isometric filling

THEOREM

Numerical optimum of this estimate:
5.38984897...

Source: Companion manuscript, Theorem 12.1 and its evaluation, lines 1510–1581; exact Lean interface in `FORMALIZATION.md`.

The method has a ceiling—and the problem has new routes

A precise limitation

The normalized *one-high-leg* resonance hierarchy, when certified by a global ambient comass bound, satisfies

$$\text{certificate} < 5.466.$$

This is not a ceiling for every nonlinear Fourier method.

Three routes beyond it

1. a Fourier–Züst profile-adapted hybrid;
2. a de Sitter comparison inequality for profile geometry;
3. genuinely multi-high-leg, low-mode corrections.

The remaining numerical gap

$$2\pi - 5.38982446 \approx 0.89336.$$

Source: Companion manuscript, Sections 13–16 and 18, especially lines 1582–1988 and 2616–2676.

What is verified, what is not

VERIFIED

Both headline surface-area bounds at their explicit Riemannian interfaces; the Fourier algebra, topological coverage, area certificates, and rigorous endpoint numerics used in those closures.

PARTIAL

Several broader standalone manuscript formulations, including general Hilbert-valued Stokes/comass statements and later resonance/operator bridges.

OPEN

Gromov's sharp 2π conjecture, the full nonlinear frontier, and any definitive priority judgment.

Kernel verification checks exact formal statements; it does not turn a stronger nearby sentence into a theorem.

6,677 PROJECT DECLARATIONS AUDITED

NO PROJECT PROOF ESCAPES

Source: Repository FORMALIZATION.md, complete 27-item ledger and trust-evidence section at commit f166e04.

Three ideas to take home

1. The no-shortcuts condition converts an unknown surface into distance profiles with exact boundary data.
2. Fourier modes share one Jacobian budget; topology turns their boundary curves into planar area certificates.
3. Nonlinear phase resonance gains because boundary action moves at first order while the comass is stationary.

The best bound here is 5.38982446; the conjectural 2π remains open.

Questions?

Backup: the metric contract in one line

Let $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ with

$$d_{\mathbb{T}}(s, t) = \min_{k \in \mathbb{Z}} |s - t + 2\pi k|.$$

If $\gamma : \mathbb{T} \rightarrow \partial M$ is arclength parametrized, traveling on the boundary gives

$$d_M(\gamma(s), \gamma(t)) \leq d_{\mathbb{T}}(s, t).$$

Thus “isometric filling” is exactly the reverse inequality:

$$d_M(\gamma(s), \gamma(t)) \geq d_{\mathbb{T}}(s, t).$$

Scaling

Circumference 2π is a normalization. For boundary length ℓ , all area bounds scale by $(\ell/2\pi)^2$.

Backup: the triangle-wave coefficient

On $[-\pi, \pi]$, the centered boundary profile is $q(t) = |t| - \pi/2$. For $n \geq 1$,

$$\frac{1}{\pi} \int_{-\pi}^{\pi} |t| \cos(nt) \, dt = \frac{2}{\pi n^2} ((-1)^n - 1).$$

Therefore

$$\hat{q}(n) = \begin{cases} -4/(\pi n^2), & n \text{ odd}, \\ 0, & n \text{ even}. \end{cases}$$

Translating the triangle wave by s contributes the phase e^{ins} , giving $z_n(\gamma(s)) = -(4/(\pi n^2))e^{ins}$.

Source: Companion manuscript, Lemma 4.1.

Backup: from Bessel energy to a Jacobian budget

At a regular point choose an orthonormal basis (e_1, e_2) and write

$$f(\theta) = da_\theta(e_1), \quad g(\theta) = da_\theta(e_2).$$

Bessel's inequality for the selected Fourier modes and orthogonality of U give

$$\sum_j (|dG_j(e_1)|^2 + |dG_j(e_2)|^2) \leq \frac{1}{\pi} \int_0^{2\pi} (f^2 + g^2) d\theta.$$

Since $J_2 L \leq \frac{1}{2}(|Le_1|^2 + |Le_2|^2)$,

$$\sum_j J_2 G_j \leq \frac{1}{2\pi} \int |\nabla a_\theta|^2 d\theta = 1 - \frac{1}{2\pi} \int |\nabla \sigma_\theta|^2 d\theta.$$

Source: Companion manuscript, Proposition 3.2.

Backup: the Givens rotations are explicit

For $m_k = 2k - 1$ and $w_k = 1/m_k$, set

$$c_k = (1 + 4w_k^2)^{-1/2}, \quad s_k = 2w_k(1 + 4w_k^2)^{-1/2}.$$

Let R_k rotate only the $(1, k)$ coordinate plane by

$$\begin{pmatrix} c_k & -s_k \\ s_k & c_k \end{pmatrix}, \quad U_N = R_2 R_3 \cdots R_N.$$

The estimates

$$c_k \geq e^{-2w_k^2}, \quad \sum_{k \geq 2} \frac{1}{(2k - 1)^2} < \frac{1}{4}$$

yield the rowwise first-harmonic dominance required in the main talk.

Source: Companion manuscript, Lemmas 5.1–5.2.

Backup: why first-harmonic dominance gives a Jordan curve

Suppose

$$z(s) = \sum_{k=1}^N a_k e^{im_k s}, \quad |a_1| > \sum_{k=2}^N m_k |a_k|.$$

Since $|e^{ims} - e^{imt}| \leq m|e^{is} - e^{it}|$,

$$|z(s) - z(t)| \geq \left(|a_1| - \sum_{k \geq 2} m_k |a_k| \right) |e^{is} - e^{it}| > 0.$$

Hence z is embedded. Straight-line homotopy to $a_1 e^{is}$ avoids the origin, so its winding is $+1$, and Green's formula gives

$$|\Omega_z| = \pi \sum_{k=1}^N m_k |a_k|^2.$$

Backup: the mod-2 coverage contradiction

Let $G : M \rightarrow \mathbb{R}^2$ be continuous and suppose $G|_{\partial M}$ is a Jordan curve C . If $y \in \Omega_C$ were omitted by $G(M)$, radial projection about y would produce

$$r_y \circ G : M \longrightarrow S^1.$$

Its boundary restriction has odd degree because y lies inside C .

But over $\mathbb{Z}_2$, the boundary class of a compact surface is the boundary of the relative fundamental class:

$$[\partial M] = \partial[M, \partial M]_2.$$

It must therefore map to zero in $H_1(S^1; \mathbb{Z}_2)$ —a contradiction. Thus $\Omega_C \subseteq G(M)$.

Source: Companion manuscript, Lemma 5.4.

Backup: the finite certificate telescopes conceptually

With $\rho_m = 4/(\pi m^2)$, the j th mixed boundary curve encloses

$$|\Omega_j| = \pi \sum_{k=1}^N m_k \rho_{m_k}^2 u_{jk}^2.$$

Sum in j and use the column identity $\sum_j u_{jk}^2 = 1$:

$$\begin{aligned} \sum_j |\Omega_j| &= \pi \sum_k m_k \rho_{m_k}^2 \sum_j u_{jk}^2 \\ &= \frac{16}{\pi} \sum_{k=1}^N \frac{1}{m_k^3}. \end{aligned}$$

The topology supplies each $|\Omega_j|$; orthogonality makes all dependence on the mixing matrix disappear after summation.

Backup: calibration and comass vocabulary

For a two-form ω , its comass on the admissible unit tangent planes is

$$\|\omega\|_{\text{comass}} = \sup_{(e_1, e_2)} |\omega(e_1, e_2)|.$$

If $\omega = d\alpha$, Stokes and the area bound give

$$\left| \int_{\partial M} G^* \alpha \right| = \left| \int_M G^* \omega \right| \leq \|\omega\|_{\text{comass}} \text{Area}(M).$$

Therefore every such construction yields the certificate

$$\text{Area}(M) \geq \frac{\text{boundary action}}{\text{global comass bound}}.$$

The nonlinear problem is an optimization of this quotient.

Backup: exact boundary action of the resonance

The perturbed boundary radius is

$$\rho_n + \lambda \rho_1^2 \rho_{n+2}, \quad \rho_n = \frac{4}{\pi n^2}.$$

Hence

$$\mathcal{B}(\lambda) = \pi \sum_{\substack{n>0 \\ n \text{ odd}}} n (\rho_n + \lambda \rho_1^2 \rho_{n+2})^2 = C_0 + b\lambda + d\lambda^2,$$

where

$$C_0 = \frac{14\zeta(3)}{\pi}, \quad b = \frac{512}{\pi^3} \left(\frac{3}{4} - \frac{\pi^2}{16} \right) > 0,$$

$$d = \frac{4096}{\pi^5} \left(\frac{7}{8}\zeta(3) + 1 - \frac{\pi^4}{48} \right) > 0.$$

Source: Companion manuscript, Proposition 9.1.

Backup: quantitative comass control

If $\delta = 1 - |\Omega_0|$ measures the deficit from linear saturation, the first and second variations obey

$$|L| \leq C_* \sqrt{\delta} + D_* \delta, \quad |Q| \leq Q_*.$$

Optimizing over $\delta \in [0, 1]$ gives, for $0 \leq \lambda < \pi^2/32$,

$$\|\Omega_\lambda\|_{\text{comass}} \leq 1 + \lambda^2 \left[Q_* + \frac{C_*^2}{4(1 - D_* \lambda)} \right].$$

Here

$$C_* = \frac{8\sqrt{2}}{3}, \quad D_* = \frac{32}{\pi^2}, \quad Q_* = \frac{128}{9\pi^2} + \frac{1280}{9\pi^4}.$$

Source: Companion manuscript, Proposition 11.1.

Backup: certified arithmetic at $\lambda = 0.03$

The one-parameter certificate evaluates to

$$\frac{\mathcal{B}(0.03)}{\mathcal{C}(0.03)} = \frac{5.4230028938\dots}{1.0061557533\dots} = 5.3898244639\dots$$

Thus the safely rounded theorem endpoint is

$$\boxed{\text{Area}(M) > 5.38982446.}$$

Numerically optimizing the explicit lower-bound formula gives

$$\lambda \approx 0.029227, \quad \text{Area}(M) \geq 5.3898489672\dots$$

Status distinction

The strict 5.38982446 endpoint is the displayed verified theorem; the numerical optimum explains the formula but is not substituted for it.

Backup: the one-high-leg spectral ceiling

For a normalized finite hierarchy, the unavoidable high-mode test gives

$$\mathfrak{C}_J(\mu) \geq 1 + \sum_{j=1}^J \mu_j^2.$$

Meanwhile the boundary action is the quadratic form $x^T M^{(J)} x$, with

$$M_{jk} = \frac{16}{\pi} \sum_{\substack{n>0 \\ n \text{ odd}}} \frac{n}{(n+2j)^2(n+2k)^2}.$$

Therefore every such certificate is bounded by $\lambda_{\max}(M^{(J)})$. The manuscript proves the infinite-operator bound

$$\|M\| < 5.466.$$

The formalization ledger records the finite algebra and trace arithmetic as verified pieces, while the full operator/comass bridge remains partial or open.

Source: Companion manuscript, Section 14; FORMALIZATION.md, Theorem 14.2 through Theorem 14.4 rows.

Backup: rescaling to any circumference

If the boundary circumference is ℓ , rescale lengths by $2\pi/\ell$. Areas scale quadratically, so the orientation-free theorem becomes

$$\text{Area}(M) \geq \frac{7\zeta(3)}{2\pi^3} \ell^2.$$

The oriented nonlinear bound scales by the same factor:

$$\text{Area}(M) > 5.38982446 \left(\frac{\ell}{2\pi} \right)^2.$$

The hemisphere benchmark becomes

$$2\pi \left(\frac{\ell}{2\pi} \right)^2 = \frac{\ell^2}{2\pi}.$$

Backup: exact formalization boundary

Item	Repository status at f166e04
Universal Fourier bound	VERIFIED at the exact compact connected Riemannian filling interface recorded in the ledger.
Explicit nonlinear improvement	VERIFIED at the project-defined conventional orientation interface; strict decimal endpoint and all-parameter formula.
Core ingredients	Fourier algebra, Givens mixing, degree/Jordan coverage, intrinsic area inequality, certificate logic, and rational numerics are verified in the headline closures.
Broader formulations	General raw-distance eikonal, generic Hilbert-valued Stokes/comass, and several later resonance/operator items are partial or open as standalone manuscript claims.
Trust receipt	GitHub Actions run 33504856733; 6,677 project declarations audited; allowed axioms exactly propext, Classical.choice, and Quot.sound.

No repository statement proves the conjectural 2π bound.

Source: Repository FORMALIZATION.md, frozen blob listed in the slide README.

References I: the conjecture and known cases

M. Gromov (1983), “Filling Riemannian manifolds,” *J. Differential Geometry* **18**, 1–147.

[DOI:10.4310/jdg/1214509283](https://doi.org/10.4310/jdg/1214509283) | [MR697984](#)

V. Bangert, C. Croke, S. Ivanov, and M. Katz (2005), “Filling area conjecture and ovalless real hyperelliptic surfaces,” *Geom. Funct. Anal.* **15**, 577–597.

[DOI:10.1007/s00039-005-0517-8](https://doi.org/10.1007/s00039-005-0517-8) | [MR2221144](#) | [arXiv:math/0405583](#)

These primary sources support the conjecture statement and the disk/genus-one status summary used in the core deck.

References II: recent bounds

R. Züst (2025), “The Riemannian hemisphere is almost calibrated in the injective hull of its boundary,” *Calc. Var. PDE* **64**, Art. 297.

[DOI:10.1007/s00526-025-03133-z](https://doi.org/10.1007/s00526-025-03133-z) | [arXiv:2104.04498](https://arxiv.org/abs/2104.04498)

J. Briggs and C. Wells (2026), “A discrete view of Gromov’s filling area conjecture.”

[arXiv:2602.17859](https://arxiv.org/abs/2602.17859)

L. Chen, X. Li, and Y. Zhong (2026), “Fourier and resonant nonlinear bounds for Gromov’s filling area problem,” companion manuscript.

[private source snapshot](#) | [formalization ledger](#)

The repository is private. These last two links require collaborator access and are not an authorization to redistribute the source.